IMPULSE V1: Erratum and Technical Correction

A methodological reassessment of TLE-based passive orbital sensing
and its implications for debris-environment model discrimination

Swayam Sahoo
Chelmsford High School
November 2026

A formal correction to the September 2026 IMPULSE V1 research report, accompanying the corrected V1.1 edition. This correction is author-initiated; in journal usage such documents are often termed corrigenda. Section and page references of the form "V1, §x, p. y" use the section numbers and printed page numbers of the original report.

1 Publication Information

Item	Details
Original report	*IMPULSE V1: Quantifying Systematic Limits of TLE-Based Passive Orbital Sensing for Microdebris Environment Model Discrimination*. Swayam Sahoo, Chelmsford High School, September 2026 (25 pp., 9 sections). Archived unmodified.
Corrected report	*IMPULSE: Quantifying the Real-Data Limits of Passive Orbital Sensing for Sub-Catalog Debris-Environment Model Discrimination* (V1.1). Swayam Sahoo, Chelmsford High School, October 2026 (25 pp., 8 sections). Current edition.
This document	Erratum and technical correction to the original report. November 2026.
Availability	Pipeline code and analysis scripts are hosted on GitHub and available from the author upon request.

2 Summary of Correction

This document provides a formal methodological correction to IMPULSE V1, originally completed in September 2026. Following the completion of V1, further investigation identified a significant limitation in the measurement covariance treatment used by the original detection pipeline: V1 assigned every object the same population-wide measurement covariance, taken from published TLE accuracy statistics [3] and treated as diagonal (V1, §3.1, p. 8), rather than characterizing the measurement behavior of each individual orbital history.

Subsequent investigation demonstrated that this assumption could produce substantial false-positive triggering under the operating conditions of IMPULSE. A revised per-object empirical covariance estimation procedure was developed, in which each stream's covariance is estimated from its own one-step prediction residuals, and was subjected to additional validation, including an end-to-end synthetic experiment

evaluating the TLE/SGP4 mean-element convention (V1.1, §5.1). The corrected pipeline, the re-calibrated threshold, and the revised fleet-scale analysis are presented in V1.1.

The corrections substantially change the detection sensitivity, the fleet-scale trigger statistics, and the interpretation of the original findings. V1 reported a raw trigger rate of 51.55% (8,161 of 15,830 windows), a scrambled-null rate of 19.46%, and a net excess of $5{,}080 \pm 103$ events, interpreting the residual as a 26–32% temporally coherent systematic floor that blocked model discrimination (V1, §7.3–V1, §7.4, pp. 17–18). V1.1, applying per-object covariance calibration, reports 6 raw triggers among 15,760 valid windows (0.038%) against 15.76 expected from the design noise rate — a deficit, not an excess — certifies zero debris-impact detections, and concludes that discrimination is unachieved because the survey's certified sensitivity, not a measured systematic floor, is insufficient to resolve the predicted impact population. The original conclusions concerning a 26–32% residual systematic floor are therefore superseded.

3 Description of the Error and Its Correction

What V1 implemented. V1 used a single population-wide measurement covariance R for all objects, taken from published TLE accuracy statistics and treated as diagonal (V1, §3.1, p. 8), with a full quadratic-form Mahalanobis statistic evaluated on the smoother covariance (V1, §3.4, Eq. 5, p. 8), a threshold of $\eta = 13.258$ from a 68,702-window combined confounder null (V1, §4.2, Fig. 4, p. 10), and the equinoctial state (a, h, k, p, q, λ) (V1, §3.1, Eqs. 1–2, p. 5).

Why it failed. Published TLE accuracy statistics describe the catalog average, not an individual stream. A minority of objects are far noisier than that average (V1.1, §3.2, Fig. 5); for those objects the assumed covariance is too tight, so the normalized forward–backward discrepancy inflates and the statistic crosses any threshold calibrated in the same units. The quadratic form compounded the problem: evaluated with the smoother covariance rather than the discrepancy covariance, it was never analytically standardized, and its magnitudes were only indices relative to whatever covariance was assumed. The result was a 51.55% fleet trigger rate against model predictions of 0.04–5.36%, a 19.46% scrambled-null rate, and an apparent net excess of $5{,}080 \pm 103$ events that V1 interpreted as a 26.7–32.0% systematic floor (V1, §7.2–V1, §7.4, pp. 17–18; Tables 1 and 2).

Why the original validation missed it. The synthetic validation injected noise consistent with the assumed covariance, so the assumed and actual covariances agreed by construction; the confounder nulls probed force-model error, not measurement noise; and the scrambled-residual null preserved each stream's residual amplitudes, so it could quantify incoherent noise but not a wrong noise *scale*. One early indication existed and was misread: the $\sim$15% negligible-Δv floor of the real-TLE injection sweep (V1, §5.2, p. 12) was in fact the signature of the misspecified noise model operating on real data, and it vanishes under the corrected covariance (0 of 149 anchor trials; V1.1, §5.2).

How the correction works. V1.1 estimates the operating measurement covariance R per object from the TLE stream itself: the per-channel scatter of one-step prediction residuals — taken as the median of squared residuals, so that a real impulse widens R only mildly instead of blinding the detector — becomes the operating covariance, floored per channel at the published statistics (the detector is never more confident

than the published accuracy, so the original error cannot recur in the opposite direction), capped at $100\times$ for pathological objects, and applied to the forward, backward, and initial covariances (V1.1, §3.2). The quadratic form is replaced by a scalar semi-major-axis channel statistic, the channel carrying the largest persistent impulse signature, with the window peak as the detection statistic, calibrated empirically at a pre-specified 0.1% per-window design false-alarm rate against a pooled synthetic null of 18,582 trials carried through the same pipeline and quality cuts as the real fleet ($\eta = 417.94$; V1.1, §3.5, §4.2). At a fixed threshold, the per-object covariance alone collapses the fleet trigger rate from 51.55% to 18.75% ($2.75\times$), identifying covariance misspecification as the dominant mechanism behind the original trigger population.

Convention consistency. An end-to-end synthetic experiment separates the two candidate mechanisms (V1.1, §5.1): under the *population-wide* covariance, a synthetic control triggers 94.7% of windows (100% with secular dynamics included); under the *per-object* covariance the same control triggers 0 of 7,000 windows. The TLE/SGP4 mean-element convention mismatch — a near-constant ≈2.9 km offset, the SGP4 representational floor — is therefore benign under the adopted covariance; the covariance misspecification is the operative error. The two effects were investigated separately and should not be conflated.

Not a controlled experiment. V1 → V1.1 also changed the state representation (equinoctial (a, h, k, p, q, λ) to (a, h, k, i, Ω, M)), the window accounting (a symmetric exclusion of 70 divergent windows, 15,830 to 15,760, applied identically to the fleet and every null population), the null construction (scrambled-spline to residual-permutation [7]), and the reference model (ORDEM 3.0 to 3.1 [8]). The numerical differences below should therefore not be read as a controlled before-and-after isolating the covariance correction alone; the fixed-threshold ablation above isolates the dominant mechanism.

4 Impact on Original Results

Table 1 summarizes the affected quantities, and Table 2 states the status of each original finding. All V1 quantitative results that depend on the measurement covariance, the detection statistic, or the threshold calibration are superseded.

Table 1: Key quantities: original report versus corrected report.

Quantity	**V1 (Sept. 2026)**	**V1.1 (Oct. 2026)**
Measurement covariance R	Population-wide published statistics, diagonal	Per-object empirical, floored and capped
Detection statistic	9-state quadratic form (Eq. 5)	Scalar semi-major-axis channel
Threshold η	13.258 (99.97th pct., 68,702-window confounder null)	417.94 (99.9th pct., 18,582-trial pooled synthetic null)
State representation	(a, h, k, p, q, λ)	(a, h, k, i, Ω, M)
Valid windows	15,830	15,760 (70 divergent excluded)
Δv_{50}, synthetic	0.041 m/s	1.19 m/s
Δv_{50}, real TLEs	0.0385 m/s	1.21 m/s
Negligible-Δv real-TLE floor	$\sim$15%	0 of 149 anchor trials
Fleet raw triggers	8,161 (51.55%)	6 (0.038%)
Realistic null	Scrambled-spline, 19.46%	Residual-permutation, $p_{99.97} = 370.7 < \eta$
Expected noise triggers	— (not tabulated)	15.76
Net excess	$5{,}080 \pm 103$ (32.09%)	None certifiable ($P \approx 0.005$ deficit)
Predicted detectable events	ORDEM 3.0: 847.9; MASTER: 6.1	ORDEM 3.1: 0.005; MASTER: 0.011 (1–10 cm)
Detection horizon	$\sim$mm–cm scale implied	$\approx$2–3 cm at 10 km/s, head-on

Table 2: Status of the principal V1 findings and the reason for each correction.

V1 finding (location)	Status	Reason
51.55% raw trigger rate (V1, §7.2, p. 17)	Superseded	Corrected pipeline: 6 of 15,760 windows (0.038%); the original rate was generated by measurement-covariance misspecification (fixed-threshold ablation: 51.55% $\rightarrow$ 18.75%).
19.46% scrambled-null rate (V1, §6.2, p. 14)	Superseded	Replaced by a residual-permutation null whose 99.97th percentile (370.7) lies below η; the permuted false-alarm rate at η is bounded below 0.03% per window.
Net excess $5{,}080 \pm 103$, 32.09% (V1, §7.3, p. 17)	No longer supported	6 observed triggers against 15.76 expected ($P \approx 0.005$); no positive excess is certifiable against either null.
26–32% systematic floor (V1, §7.3–V1, §7.4, pp. 17–18)	Superseded	The trigger population interpreted as a coherent systematic floor was dominated by covariance misspecification, which the corrected calibration removes or absorbs into the empirically calibrated threshold; the corrected analysis finds no excess to decompose.
$\Delta v_{50} = 0.041$ / 0.0385 m/s (V1, §5.1–V1, §5.2, pp. 11–12)	Revised	The statistic is rescaled by the correctly estimated per-object covariance; corrected values are 1.19 m/s (synthetic) and 1.21 m/s (real TLEs).
ORDEM 847.9 / MASTER 6.1 detectable events (V1, §5.3, p. 13)	Revised	Corrected sensitivity moves the detection horizon to $\approx$2–3 cm; corrected predictions are 0.005 / 0.011 detectable events in the 1–10 cm band and none in the 1 mm–1 cm band (ORDEM 3.1).
Discrimination impossible due to a large residual systematic floor (V1, §7.4, §8–§9, pp. 18–20)	Revised	Discrimination remains unachieved, but because the certified sensitivity cannot resolve a predicted population of $\lesssim$0.011 detectable events — not because of a measured systematic floor. The corrected result is a weak upper bound ($\approx$280–570$\times$ the model-predicted rate), not a floor measurement.

5 Revised Scientific Conclusions

V1 concluded that the detection architecture was sensitive but that a 26–32% temporally coherent systematic floor in real TLE histories constituted an information limit blocking ORDEM/MASTER discrimination, and it set the decomposition of that floor as the objective for the next iteration. The corrected analysis revises both the result and the cause: no positive excess is certifiable against either background, no trigger candidate

is attributable to debris, and the survey certifies zero debris-impact detections — the absence of a certifiable excess, *not* the absence of impacts (Tables 1 and 2). In place of a floor measurement, the corrected result is a weak one-sided upper limit on the $\geq$2–3 cm impact rate.

The discrimination verdict is unchanged in outcome and revised in cause. The certified impulse floor (Table 1) leaves both models predicting well under one detectable event over the present exposure, and the two models agree within a factor of $\approx$3 in the decision-relevant 1–10 cm band, so even a positive excess could not have discriminated between them at this sensitivity. V1 attributed the failure to a measured systematic floor; V1.1 attributes it to sensitivity, and the path forward is correspondingly revised: longer exposure, a larger fleet, and a lower Δv_{50} — not decomposition of a floor the corrected analysis does not find.

6 Scope of the Correction and Remaining Limitations

What remains valid. The fleet construction and data set, the dual-EKF + Fraser-Potter + FOGM architecture, the motivation for nonsingular (h, k) elements, the research framing, the qualitative finding that force-model confounders do not generate triggers under a correctly specified covariance, and the negative model-discrimination verdict in outcome all remain sound.

What is superseded. Every V1 quantitative result that depends on the measurement covariance, the detection statistic, or the threshold calibration, as itemized in Tables 1 and 2.

What remains unresolved. The six corrected-pipeline trigger candidates remain unattributed (five lie within the calibrated null tail; one, NORAD 31949, lies above the null maximum). The corrected survey carries its own residual limitations, which V1.1 documents in full (§7.2); none of them restores a measurable debris signal, and they bound what the corrected survey can claim.

7 References and Related Publications

The document set comprises the original report (archived, unmodified), this erratum, and the corrected V1.1 edition (current). V1 should continue to be cited as the original report, with V1.1 and this erratum cited for all corrected results.

References

[1] Sahoo, S. (2026). *IMPULSE V1: Quantifying Systematic Limits of TLE-Based Passive Orbital Sensing for Microdebris Environment Model Discrimination*. Research report, Chelmsford High School, September 2026. Original report; archived unmodified.

[2] Sahoo, S. (2026). *IMPULSE: Quantifying the Real-Data Limits of Passive Orbital Sensing for Sub-Catalog Debris-Environment Model Discrimination* (V1.1). Research report, Chelmsford High School, October 2026. Corrected edition; current.

[3] Osweiler, V. P. (2006). *Covariance estimation and autocorrelation of NORAD two-line element sets* (Publication No. AFIT-GSS-ENY-06-M09) [Master's thesis]. Air Force Institute of Technology. Source of the population-wide covariance used in V1; retained in V1.1 as the per-channel floor.

[4] Bennett, A. A., Carpenter, R., & Schaub, H. (2023). Methods to detect impact-induced orbit perturbations using spacecraft navigation data. *Journal of Spacecraft and Rockets*, 60(4), 1043–1062.

[5] Hoots, F. R., & Roehrich, R. L. (1980). *Models for propagation of NORAD element sets* (Spacetrack Report No. 3). U.S. Air Force Aerospace Defense Command.

[6] Vallado, D. A., & Cefola, P. J. (2012). Two-line element sets — practice and use. *Proceedings of the 63rd International Astronautical Congress*, (IAC-12-C1.6.12).

[7] Ernst, M. D. (2004). Permutation methods: A basis for exact inference. *Statistical Science*, 19(4), 676–685.

[8] Matney, M., Manis, A., Anz-Meador, P., Gates, D., Seago, J. H., Vavrin, A., & Xu, Y.-L. (2019). The NASA Orbital Debris Engineering Model 3.1: Development, verification, and validation. *First International Orbital Debris Conference*, 2109, Article 6134.

[9] Horstmann, A., Manis, A., Braun, V., Matney, M., Vavrin, A., Gates, D., Seago, J., Anz-Meador, P., Wiedemann, C., & Lemmens, S. (2021). Flux comparison of MASTER-8 and ORDEM 3.1 modelled space debris population. *Proceedings of the 8th European Conference on Space Debris*.